

Droplet impact on cylindrical surfaces: Effects of surface wettability, initial impact velocity, and cylinder size

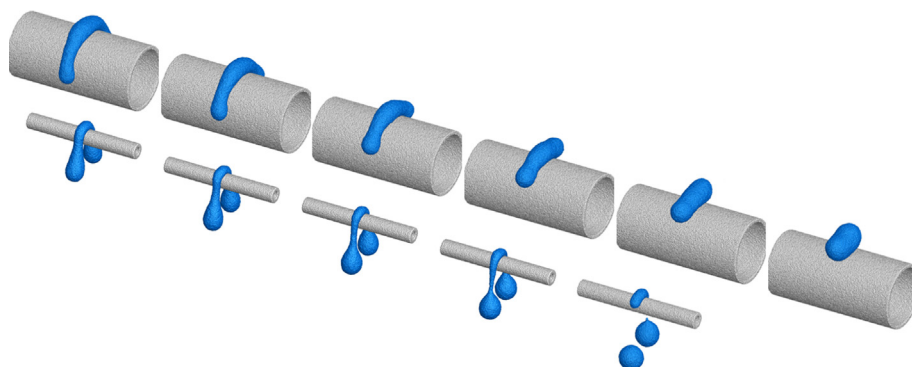
Yulei Wang^{a,b}, Yuxiang Wang^{c,*}, Shaorong Wang^{a,b}

^a State Key Laboratory of Advanced Electromagnetic Engineering and Technology, Huazhong University of Science and Technology, Wuhan, Hubei Province 430074, China

^b School of Electrical and Electronic Engineering, Huazhong University of Science and Technology, Wuhan, Hubei Province 430074, China

^c Department of Chemical Engineering, Monash University, Clayton, Victoria 3800, Australia

GRAPHICAL ABSTRACT



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ABSTRACT

Hypothesis: Droplet impact on cylindrical surfaces is a widespread phenomenon observed in both nature and industrial applications. The wettability and size of the cylinders, and the initial impact velocity, are expected to influence the impact dynamics, such as generating different droplet morphology, spreading/retracting processes, and contact time, which could finally determine the anti-icing property of power transmission lines.

Experiments: A particle-based mesh-free numerical approach, many-body dissipative particle dynamics, is employed to study the impact dynamics. The cylinders with three different sizes and three types of wettability are modelled, and different initial impact velocities are set.

Findings: The simulations show the two liquid films (or lamellae) formed after impact can coalesce at the bottom site of a hydrophilic cylinder while they keep separate on hydrophobic and superhydrophobic cylinders. The maximum spreading diameter in the axial direction increases with the increase of impact velocity and cylinder size. Besides, the distribution of droplet detachment mode shows a strong dependence on initial impact velocity and cylinder size. In bouncing mode, the contact time increases with the increase of impact velocity while in dripping/bouncing mode, it has an opposite trend. The splashing angle is also observed to be dependent on the cylinder size.

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* Corresponding author.

E-mail address: yuxiangwang123@gmail.com (Y. Wang).

1. Introduction

The droplet impact process on cylindrical surfaces is a widespread phenomenon observed in industrial processes, such as self-cleaning [1], spraying [2], printing [3], and anti-icing [4,5] processes. For the anti-icing process, one important application is the design of power transmission lines. Power transmission lines in cold regions always suffer the damage caused by the icing problem [6], which can lead to huge loss and inconvenience in people's life [7]. One promising strategy of addressing this issue is to prevent the droplet icing on the cylindrical solid surfaces after impact. The fundamental aspect of this strategy is to comprehensively understand the droplet impact dynamics on the cylindrical surfaces.

Many researchers have thus put great effort into the study of droplet impact on cylindrical objects. Some of the studies which are closely relevant to the present work are reviewed in detail below [8,12–17]. Khurana et al. [8] performed the experiments of droplet impact on hydrophilic and superhydrophobic cylindrical surfaces with different sizes, and three parameters, the wetting fraction, non-dimensional film thickness, and spreading factor were investigated. Their results show that on hydrophilic surfaces, when the cylinder has a size comparable with the droplet size, different impact velocity can cause different post-impact droplet deformation. When impact at high velocity, the droplet completely engulfs the cylinder and form two liquid films at both sides in the azimuthal direction. As the two films spread, they can meet each other from the two sides of the cylinder and coalesce at the bottom side. While when impact at low velocity, the droplet only engulfs the cylinder partially, and the two liquid films cannot meet and coalesce at the bottom side, they may meet at a point below the bottom of the cylinder, or still keep separate and form two secondary droplets, the actual situation depends on whether the impact velocity is low enough. Also, their experiments show the effect of the cylinder size, as smaller cylinders can cause the coalescence of the liquid films or even dripping of the droplet, while a large cylinder can prevent droplet spreading to the bottom of the cylinder. Their further experiments show that droplet impact on superhydrophobic surfaces can hardly induce the coalescence of the liquid films at the bottom of cylinders even with different sizes and the wetting fraction is significantly less on superhydrophobic surfaces when compared with that on hydrophilic surfaces. Superhydrophobic surfaces [9–11], due to their excellent self-cleaning and anti-icing properties, can be a good surface candidate in many applications. Zhang et al. [12] carried out experiments to study how the droplet spreading dynamics is influenced by cylindrical superhydrophobic surfaces with different diameter ratio (defined as the ratio of cylinder diameter to initial droplet diameter) ranging from 0.15 to 7.55, and the effect of different initial impact velocity. Their work discussed several interesting droplet morphologies, the droplet can deform from jug-like morphology gradually to wing-like morphology with the decrease of diameter ratio or increase of initial impact velocity. They also identified three stages during the impact process: the initial stage, from the beginning of touching the surface till reaching the maximal spreading diameter in the axial direction; the convergence stage, in which the contact line begins to retract in the axial direction while still spreads in azimuthal direction; the final stage, the contact line retracts at both directions simultaneously. They also observed the stretched breakup when the impact velocity is high or when the diameter ratio is low. The reason is mainly attributed to a combined effect of the downward inertia of liquid in two sides of the cylinder and the upward bounce of liquid at the top side. They also explained the stretched breakup by the mismatch of spreading and retracting in axial and azimuthal directions. Hung

et al. [13] also performed experiments to investigate the droplet impact on cylindrical objects and identified three different dripping modes: for disintegration mode which often occurs when the droplet has a moderate impact velocity and impacts on a cylinder with a size smaller than the drop, or impacts on a larger cylinder when the impact velocity is high, the droplet can be split into two halves with a “V” shape and then drips; for momentum-induced dripping, the droplet can wrap the whole cylinder and coalesce at the bottom side with the formation of a large fragment, this fragment detaches finally by the downward momentum; the gravity-induced dripping which usually occurs when the droplet impacts on small cylinders with low impact velocity, the downward momentum is not big enough, only when the pendent droplet grows big enough and can overcome the capillary effect, then the dripping can happen. Andrew et al. [14] conducted both experiments and simulations of droplet impact on a flat substrate with cylindrical ridges to study how to reduce the contact time. They found that when the diameter ratio is over one, the contact time can decrease slowly with the decrease of the diameter ratio; when the diameter ratio approaches one, the contact time reaches a minimal value; with further reduction in the diameter ratio, the contact time will increase. The momentum imbalance between the azimuthal and axial directions caused by the existence of curvature on the cylindrical ridges is used to explain the mechanism. For the multi-physics aspect, Jin et al. [15] considered three radii of cylinders ranging from 10 mm to 25 mm and carried out experiments of droplet impact and freezing processes on these cylindrical surfaces. Their results show that the maximum spreading diameter in azimuthal direction is always larger than that in the axial direction, but the cylinder size has very limited influence on the maximum spreading diameter in both directions. The variation of temperature and cylinder size are found to be irrelevant to the ice bead formation. Besides the experimental effort, some researchers performed numerical simulations to investigate the droplet impact on cylindrical surfaces to gain more insights [16,17]. By using a coupled level-set and volume-of-fluid method, Liu et al. [16] performed numerical simulations to investigate the droplet impact on cylindrical surfaces with different hydrophobicity and impact velocity. Their results indicate that the droplet can bounce from surfaces with higher hydrophobicity and is easy to be broken on a superhydrophobic surface. The pressure difference from outside to the inside of the droplet was found to affect the spreading, rebound, and splashing processes. By using the lattice Boltzmann method, Shen et al. [17] performed a 2D simulation of droplet impact on a round surface. The deformation of the droplet can be divided into four typical stages: moving, spreading, nucleating of liquid films at the bottom, and dripping. A critical range of Weber number can be identified in which the dripping can happen, otherwise, the droplet will coalesce at the bottom side when the Weber number is smaller or splash when the Weber number is larger. Also, wettability plays an important role in the post-impact process, as hydrophilic surfaces promote the dripping while hydrophobic surfaces promote the splashing.

Though many previous experimental and numerical studies have been performed, a comprehensive study of influences from wettability, object size, and the impact velocity is still needed, and the detailed description of the phenomena during the impact process and analysis of the underlying mechanisms are still insufficient. The present work aims to investigate how the surface wettability, the cylinder size together with the impact velocity influence the droplet dynamics, such as the morphology of droplet, spreading and retracting, bouncing and dripping, and the contact time dependence. A detailed description of the features during the dynamical process and analysis of the reasons are provided to deepen the understandings of the impact dynamics.

2. Simulation methods and modelling

Many-body dissipative particle dynamics (MDPD) is used to perform the simulations. MDPD is a particle-based mesh-free numerical method [18] which is derived from the original dissipative particle dynamics (DPD) [19,20]. Generally, there are three governing equations between a pair of DPD particles, they are conservative force, dissipative force, and random force. However, in DPD, the conservative force is purely repulsive which is suitable for the simulation of gas systems but not the liquid systems, as shown by Eq. (1). Later, MDPD was proposed to address this issue. In MDPD, the conservative force includes a local-density-based repulsive term and a position-dependent attractive term, as shown by Eq. (2). This kind of formation enables it to simulate the complex liquid systems with free liquid/gas interface, such as multi-phase flow and the droplet interaction with solid surfaces. Eq. (3) shows the dissipative force and Eq. (4) shows random force, they still keep the same forms as in DPD.

$$\mathbf{F}_{ij}^C = -A\omega^C(r_{ij})\mathbf{e}_{ij} \quad (1)$$

$$\mathbf{F}_{ij}^C = A\omega^C(r_{ij})\mathbf{e}_{ij} + B(\bar{\rho}_i + \bar{\rho}_j)\omega_d(r_{ij})\mathbf{e}_{ij} \quad (2)$$

$$\mathbf{F}_{ij}^D = -\gamma\omega^D(r_{ij})(\mathbf{e}_{ij}\hat{\mathbf{A}} \cdot \mathbf{v}_{ij})\mathbf{e}_{ij} \quad (3)$$

$$\mathbf{F}_{ij}^R = \sigma\omega^R(r_{ij})\xi_{ij}\mathbf{e}_{ij} \quad (4)$$

MDPD simulates liquid systems at mesoscale and has been widely used in the investigation of liquid flow [21–23], wetting phenomena [24–27], and droplet impact dynamics [28–30]. These applications of MDPD have proved its capacity in simulating the concerned system in this work. A recently proposed boundary condition [31] which considers the geometry of the solid wall, the relative distance between liquid particles and solid walls, and the velocity of liquid particles, is employed in the present work. This boundary condition can effectively prevent the penetration of liquid particles into solid walls, especially at high initial impact velocity. The values of parameters used in this work are mainly from our previous work [32], only the values of attractive coefficients for three kinds of wettability are different from the previous study and provided in Table 1. The wettability is represented by apparent contact angle θ_{app} , which is defined as the contact angle of a stable droplet on a flat solid surface after being gently released. All the units in the present work are still presented in MDPD reduced units and the gravitational effect is ignored due to the relatively small length scale in MDPD method.

Three cylinders are modelled, their sizes are presented by a non-dimensional ratio D^* , named diameter ratio, which is defined as the ratio of their diameters to the droplet diameter, as illustrated in Fig. 1. The cylinders are made hollow to reduce the computing cost. The droplet, with a radius of 12, is initially suspended above the cylinders, then released with an initial velocity v . Periodical boundaries are set in all directions. The gravitational effect is ignored due to the small length scale. All units in the simulations are MDPD reduced units, which are dimensionless.

Table 1
Wettability used in this work and the corresponding values of attractive coefficients.

Parameter	Symbol	Value	θ_{app}
Attractive coefficient (solid/liquid)	A_{sl}	−70.0	67°
Attractive coefficient (solid/liquid)	A_{sl}	−50.0	105°
Attractive coefficient (solid/liquid)	A_{sl}	24.5	151°

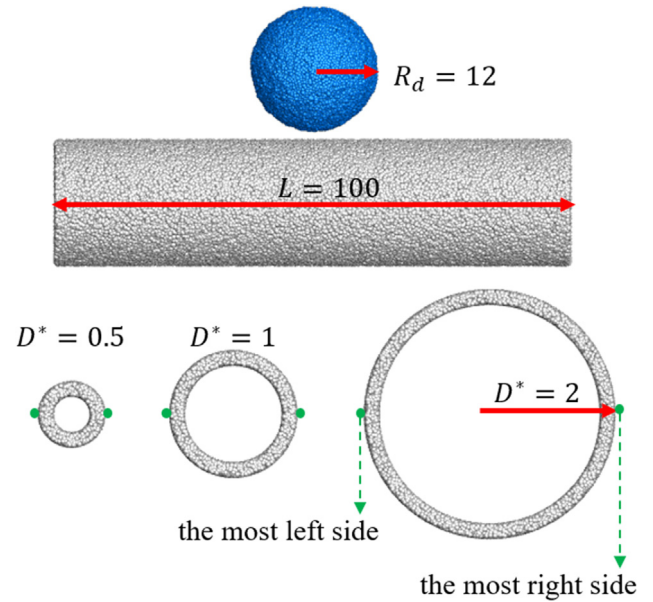


Fig. 1. Illustrations of the droplet and the three cylinders. All cylinders are with the same length $L = 100$. The green points on the cylinders indicate the most left and right sides of the surfaces, which will be discussed in the next section. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3. Results and discussions

3.1. Wettability

The snapshots of the post-impact droplet are presented in a continuous-time series in Fig. 2 to illustrate the effect of different wettability on the droplet morphology. For a fixed diameter ratio $D^* = 0.5$ and impact velocity $v = 5$, the droplet deforms as a strong function of the wettability. For impact on the hydrophilic cylinder with $\theta_{app} = 67^\circ$, the contact line can quickly spread on the cylinder in both axial and azimuthal directions. When it spreads to the most left and right sides of the cylinder, two liquid lamellae appear (see $t = 4.4$ in $\theta_{app} = 67^\circ$ column, Fig. 2). Due to the high initial impact velocity and the curved surface of the cylinder, the lamellae don't flow totally along with the contact lines which only progress along the cylinder surface after the initial impact. Instead, they flow a little bit outwards compared with the contact lines flowing along the curved surface (see $t = 9.4$ in $\theta_{app} = 67^\circ$ column, Fig. 2). Later, because of the hydrophilic property of the cylinder, the contact lines keep wetting the surface and spreading towards the bottom side. A pulling effect, which originates from the continuously spreading of the contact lines and the viscosity of the liquid, is generated and applied to the two lamellae. They are thus pulled to meet each other and coalesce at a point below the bottom of the cylinder (as indicated by the red arrows at $t = 19.4$ snapshots in $\theta_{app} = 67^\circ$ column, Fig. 2). When the two lamellae coalesce, due to the surface tension of the liquid and the downward inertia of the lamellae, their morphology shrinks from the lamella shape into two liquid pillars with bulged ends, or termed as pendant secondary droplets, as shown by the snapshots at $t = 39.4$. It is interesting to notice that though the two lamellae shrink into a pendant droplet, the contact line in the axial direction doesn't shrink or retract, as the side view of the snapshot shows. This fact indicates a strong capillary effect: the hydrophilicity of the cylinder surface lets the liquid prefer to stay at the surface and stick on it, while the surface tension of the liquid in the air lets the lamellae shrink into

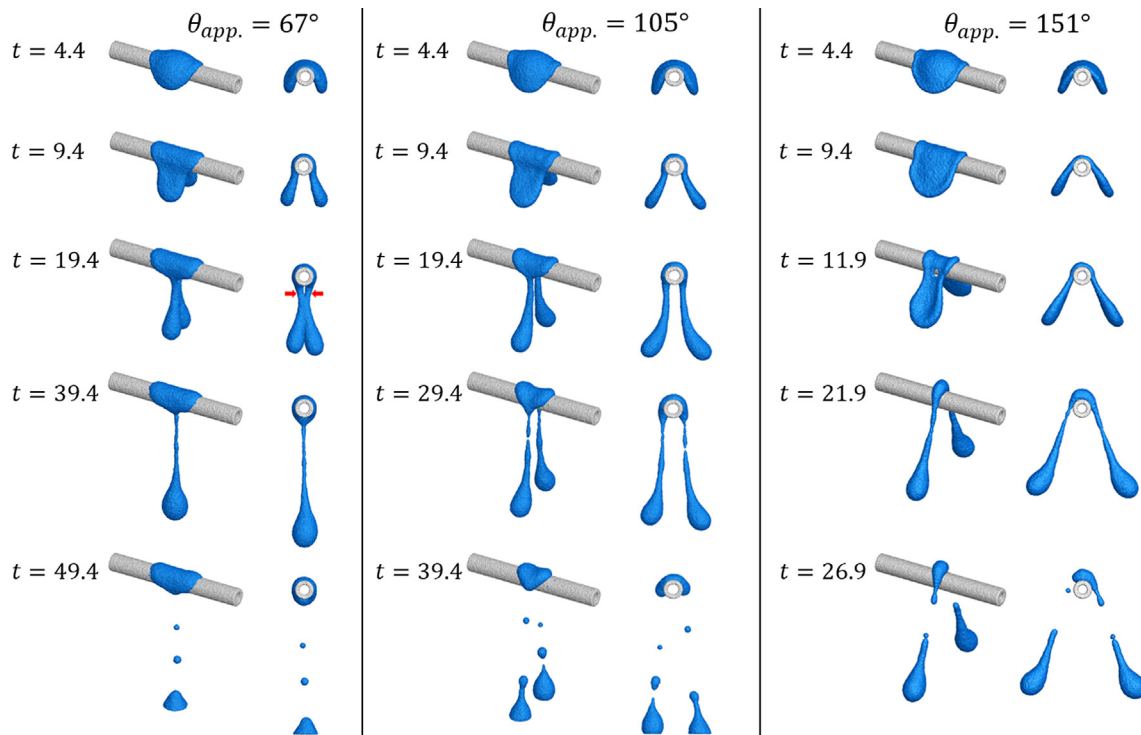


Fig. 2. The side and axis views of the morphology variation of the post-impact droplet on cylinders with different wettability, the diameter ratio is $D^* = 0.5$ and the initial impact velocity is $v = 5$.

a liquid pillar with a bulged end, in other words, both of the liquid/solid and liquid/air interfaces are trying to find their energy minima in the process. Snapshots at $t = 39.4$ also show that the bulging end has grown into a secondary droplet and keeps sagging. The liquid pillar which connects the secondary droplet with the cylinder is becoming thinner and thinner and eventually breaks up into two tiny droplets due to the continuous necking, and then the secondary droplet drips from the cylinder, as shown by snapshots at $t = 49.4$ in $\theta_{app} = 67^\circ$ row. This dripping process with a single liquid pillar can also be frequently observed in experiments with similar impact conditions [8,13,33,34], such as in Khurana and co-workers' experiment, the droplet impacts on the hydrophilic surface with $\theta_{app} = 45^\circ$ at high impact velocity. The dripping in the present simulation can also be seen as a momentum-induced dripping mode described in Hung and co-workers' experiments [13].

When the droplet impacts on the hydrophobic cylinder with $\theta_{app} = 105^\circ$, at the very beginning of the spreading, the droplet looks almost the same as impact on the hydrophilic cylinder with $\theta_{app} = 67^\circ$, as the first-line and second-line snapshots in $\theta_{app} = 105^\circ$ column in Fig. 2 show. The difference starts from the moment $t = 19.4$, as the contact line in azimuthal direction continuously spreads, the contact line in axial direction witnesses some retraction. The snapshot shows that though the contact line in azimuthal direction still spreads around the cylinder, it doesn't go so far as that on the hydrophilic surface. Moreover, the continuous contact line spreading stops as snapshots at $t = 29.4$ show. Thus, there is no pulling effect brought by the continuous spreading contact lines, and therefore the two lamellae cannot meet and coalesce during the period when the contact line in the axial direction is shrinking and the secondary droplets are sagging. Instead, the two liquid pillars keep a vertical direction with two secondary droplets at the ends. With the sagging of the two secondary droplets, the liquid pillars become thinner and unstable. Finally, the liquid pillars cannot support the secondary droplets and break up into

several tiny droplets, as shown by snapshots at $t = 39.4$. The secondary droplets also drip almost vertically. After the dripping, the contact lines in both directions retract back to some extent. This phenomenon indicates that compared to the contact line spreading in azimuthal direction on a hydrophilic surface, the spreading of contact line on this hydrophobic surface is not a spontaneous process, it is driven by the downward sagging of the lamellae and the liquid pillars. This vertical dripping with two separated liquid pillars and recoil of contact lines are also observed in Khurana and co-workers' experiment [8].

When releasing a droplet on a superhydrophobic cylinder with $\theta_{app} = 151^\circ$, the wettability effect can be further observed. Still, at the very beginning of the impact, the droplet shows a similar morphology with that on hydrophilic and hydrophobic surfaces, as the first-line and second-line snapshots in $\theta_{app} = 151^\circ$ column in Fig. 2 show. However, just after that, the contact line in the azimuthal direction almost stop spreading around the cylinder due to the superhydrophobicity of the surface and at the same time, the contact line in the axial direction is still retracting, leading to a smaller wetting area compared to that on hydrophilic and hydrophobic surfaces, as shown in snapshots at $t = 11.9$. The smaller wetting area is also observed in Khurana et al.'s experiments [8], in which they compared the wetting area on the superhydrophobic cylinder with that on the hydrophilic cylinder. The two liquid lamellae always keep separated with an intersection angle, making the whole droplet look like a bee wing, which is also described in experiments [8,12]. The bee-wing shape is always kept even when the lamellae shrink into two single liquid pillars, as the snapshots at $t = 21.9$ show. The recoil of the contact lines in both directions is more obvious compared with that on the hydrophobic surface, and the liquid has accumulated in the original impact site and makes the liquid thickness on the top site of the cylinder increase significantly. The ongoing recoil of the contact lines together with the sagging of the two secondary droplets makes the liquid pillars even more unstable, especially at the site near the retracting

contact line in the azimuthal direction. Later, the two liquid pillars are stretched to break up, leading to the dripping of the two secondary droplets. It is of interest to notice that even after the dripping, the falling secondary droplets, tiny droplets, and segments of the liquid pillars still keep the same intersection angle. This is due to the strong repellent effect of the superhydrophobic surface which not only cannot provide a pulling effect to reduce the intersection angle, it even repels the further wetting of the contact lines and pushes the two liquid pillars to stay away from each other. This stretched breakup on the superhydrophobic surface is also observed by Zhang et al. [12].

A similar characteristic for droplet impact on all three surfaces is that, at the very beginning of the impact, the droplet morphology is almost identical, as the first-line and second-line snapshots in the three columns show. This stage can be seen as an inertia stage, as the impact velocity rather than the surface properties dominates the droplet deformation. After the inertia stage, the droplet morphology during the spreading and dripping processes begins to show significant differences, which indicates that the processes begin to be governed by the wettability. This is a capillary stage, as the contact line in azimuthal direction can fully wrap the cylinder on the hydrophilic surface and coalesce at the bottom while the droplet can only partially wet the hydrophobic and superhydrophobic cylinders in azimuthal direction with the liquid lamellae shrinking into two separated liquid pillars with secondary droplets. The two liquid pillars can be vertically parallel or keep a symmetrical bee-wing shape, depending on the strength of hydrophobicity of the cylinder surfaces. Also, the more hydrophobic the cylinder is, the closer the necking points will be to the cylinder. That is because the recoil of the contact lines can accelerate the necking process near the retracting contact lines, and the recoil is quicker on the superhydrophobic surface than that on the hydrophobic surface.

The effect of wettability on the droplet impact can also be indicated by wrapping angle and non-dimensional film thickness (NFT) on the top site of the cylinder. The wrapping angle θ_w , defined as the angle of the frontier points of the contact lines to the center point of the cylinder, as shown in the snapshot in Fig. 3(a), can be used to indicate the extent of the spreading in azimuthal direction; the NFT on the top site h^* , defined as the ratio of the thickness of the film on the top side of the cylinder to the diameter of the initial droplet, as shown in Fig. 3(b), can be used to show the liquid downward drainage and upward recovery. Both Fig. 3(a) and (b) show a very significant influence of the wettability difference.

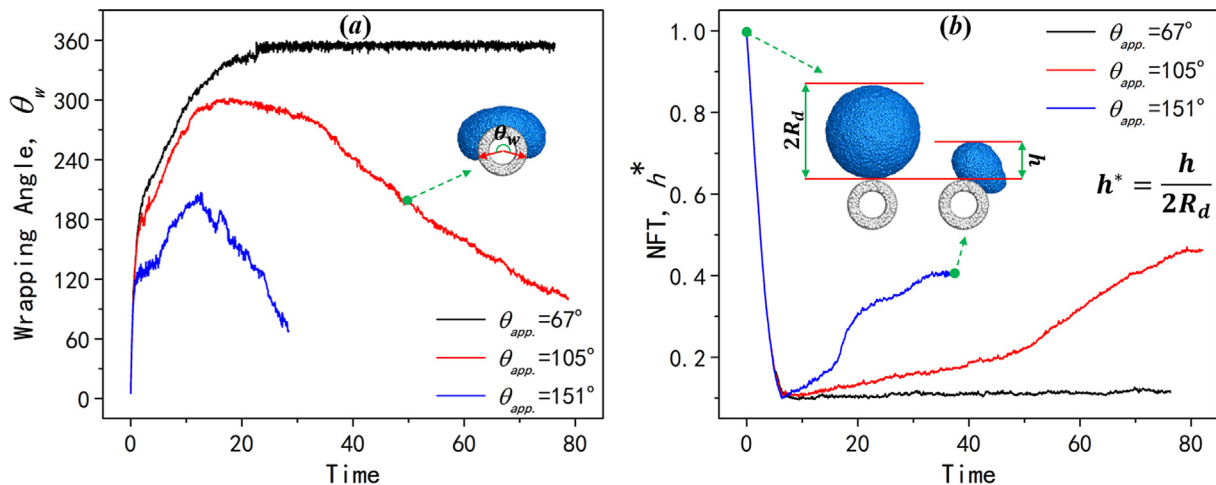


Fig. 3. The time-dependent variation of (a) wrapping angle θ_w and (b) non-dimensional film thickness (NFT) h^* on the top site of the cylinder. The diameter ratio is $D^* = 0.5$ and the initial impact velocity is $v = 5$.

From Fig. 3(a) we can observe that, on the hydrophilic cylinder, the droplet finally wraps the whole cylinder, indicated by $\theta_w = 360^\circ$ after a slower increase in wrapping angle compared to the initial fast increase. For the hydrophobic cylinder, the droplet first wraps most of the cylinder but not all, as the maximum wrapping angle is almost 300° . However, after reaching the maximum value, the wrapping angle starts a slow decrease, indicating the recoil of the contact line in the azimuthal direction. When the droplet impact on the superhydrophobic cylinder, the droplet only wraps about half of the cylinder, and then witnesses a quick recoil, corresponding to the results shown by the snapshots in Fig. 2. For the NFT at the top site of the cylinders, the distinct time-dependent variation pattern of thickness also clearly shows the wettability effect. For impact on the hydrophilic cylinder, once the thickness reaches to the minimal value, it will keep at it. However, for impact on hydrophobic and superhydrophobic cylinders, the thickness will recover after reaching the minimal values, and this process will be much faster on the superhydrophobic surface.

3.2. Cylinder size and initial impact velocity

Besides the wettability, the cylinder size and initial impact velocity also have a significant influence on the droplet impact dynamics. In this section, the superhydrophobic cylinder with $\theta_{app} = 151^\circ$ is focused to investigate these influences. Though many experiments have frequently discussed these effects [8,12,13,14,16], the present will inspect the effects on droplet morphology and spreading dynamics from a wider range of initial impact velocity and three cylinder sizes which are smaller $D^* = 0.5$, identical $D^* = 1$, and larger $D^* = 2$ compared with the droplet size.

Fig. 4 shows the results of a droplet impact on the three cylinders at different velocities from $v = 0.5$ to $v = 6$. Fig. 4(a) shows the maximum non-dimensional spreading diameter (NSD). NSD is defined as the ratio of spreading length in the axial direction over the initial droplet diameter. It can be observed that the maximum NSD is a strong function of the initial impact velocity, as it increases with the increase of the initial impact velocity. When impact on cylinders with different sizes, the contact line can spread further along the axial direction on cylinders with large size, especially at higher impact velocity. This phenomenon can be explained by the amount of liquid spreading in different directions. When the cylinder size is big, the surface of it is nearly flat, the amount of

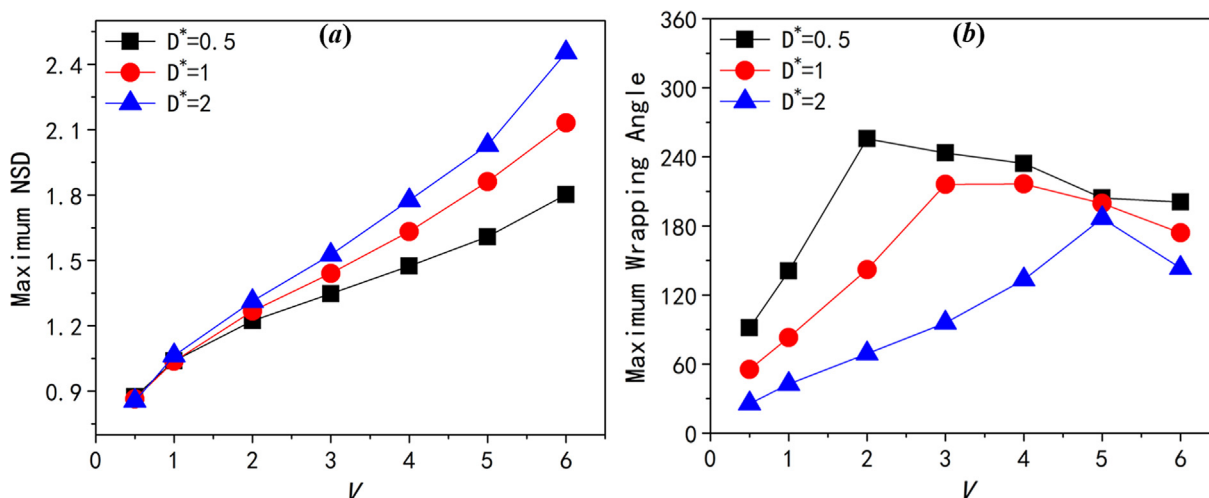


Fig. 4. Droplet impact on cylinders with diameter ratios $D^* = 0.5, 1$, and 2 at different impact velocity from $v = 0.5$ to $v = 6$. (a) shows the results of maximum non-dimensional spreading diameter (NSD), (b) shows the results of the maximum wrapping angle.

liquid spreading in axial and azimuthal directions are considerable, thus the spreading length in the two directions will also be similar. However, when the cylinder size is small, especially smaller than the droplet size, after impact, most liquid of the droplet prefers to spread in the downward direction (or azimuthal direction) since it is a vertical impact. Therefore, at the same time, the liquid which can be used for the spreading in the axial direction becomes very few, this is especially true at high impact velocity. As a result, no enough liquid supports a further spreading in the axial direction, leading to the shorter maximum NSD on small cylinders. This analysis can be further supported by the results of the maximum wrapping angle on different cylinders at different impact velocity shown in Fig. 4(b), as the maximum wrapping angles are always larger on smaller cylinders at the same impact velocity, indicating that the contact line can spread further in the azimuthal direction.

Fig. 4(b) also shows an interesting trend: for impact on all cylinders, the maximum wrapping angle always first increases and then decreases with the increase of initial impact velocity. Since impact velocity is the only variation for a single plot of wrapping angle, different initial momentum should be the main reason for this trend. For a given size of cylinder, at low impact velocity regime, the relatively low momentum cannot induce the dripping, the droplet may stick on the cylinder or bounce from it, the increase of impact velocity can force more liquid to participate into the wetting of cylinder surface, thus the contact lines can spread further in the azimuthal direction. However, at high impact velocity regime, the large momentum can induce the downward dripping, the more the dripped liquid amount is, the less the liquid will be left to support the spreading of the contact line in the azimuthal direction, finally the wrapping angle will decrease accordingly.

Fig. 5 gives the distribution of the droplet detachment modes, such as bouncing from the cylinders, dripping & bouncing from the cylinders, or sticking (fully or partially) on the cylinders. Note: the dripping & bouncing mode is named as dripping/bouncing mode in the present work. The distribution of droplet detachment modes has a good consistency with results in Fig. 4(b). For impact on the cylinder with $D^* = 0.5$, the droplet totally bounces from the cylinder at $v = 0.5 - 2$, while it will witness a dripping/bouncing mode and then totally leave the cylinder at $v = 3 - 5$. However, when increasing the impact velocity to $v = 6$, most liquid leaves the cylinder as secondary droplets by dripping, but still a little amount of liquid is observed to remain on the surface, leading to a partial sticking mode, see the impact process of it in Fig. 6. For impact on the cylinder with $D^* = 1$, at very low impact velocity

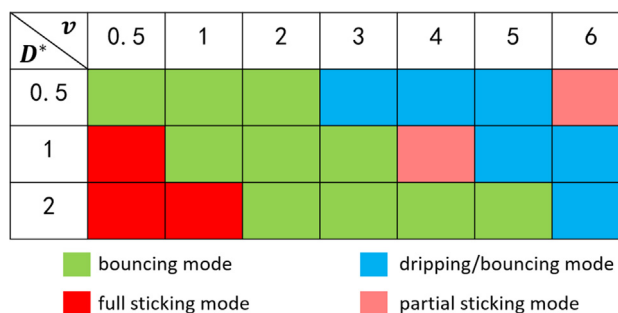


Fig. 5. The distribution of droplet detachment modes. Green color represents the bouncing mode, the blue color represents the dripping/bouncing mode, deep red represents the full sticking mode, and light red represents the partial sticking mode. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$v = 0.5$, the droplet totally sticks on the top site of the cylinder due to the insufficient upward momentum transferred from the retraction of contact lines, leading to the full sticking mode, see its impact process in Fig. 6. The droplet can have a bouncing mode from $v = 1 - 3$, and a dripping/bouncing mode from $v = 5 - 6$, with an exception of sticking mode at $v = 4$. The demonstrations of bouncing mode and dripping/bouncing mode on this cylinder can also be found in Fig. 6. However, when impact on the large cylinder with $D^* = 2$, the sticking velocity increases to $v = 1$, and the velocity range of bouncing mode increases to $v = 2 - 5$, and the dripping/bouncing mode only happens at $v = 6$.

According to the results shown in Fig. 5, more insights can be drawn. Firstly, the full detachment, either bouncing or dripping, is easier to happen on small cylinders compare with larger ones: as on the cylinder with $D^* = 0.5$, 6 impact velocities, from $v = 0.5 - 5$, can trigger the full detachment, while only 5 velocities can cause the full detachment on larger cylinders. When gradually increase the cylinder size, the surface at the top of the cylinder becomes more and flatter, as a result, the full sticking will be easy to happen at low impact velocity regime. Then, for all kinds of cylinder size, dripping/bouncing mode always happens after bouncing mode with the increase of impact velocity. This phenomenon can be explained by the functions of spreading in different directions on curved surfaces. The spreading in the axial direction is a storing process of surface energy, and after spreading, the stored surface energy will be transferred into an upward

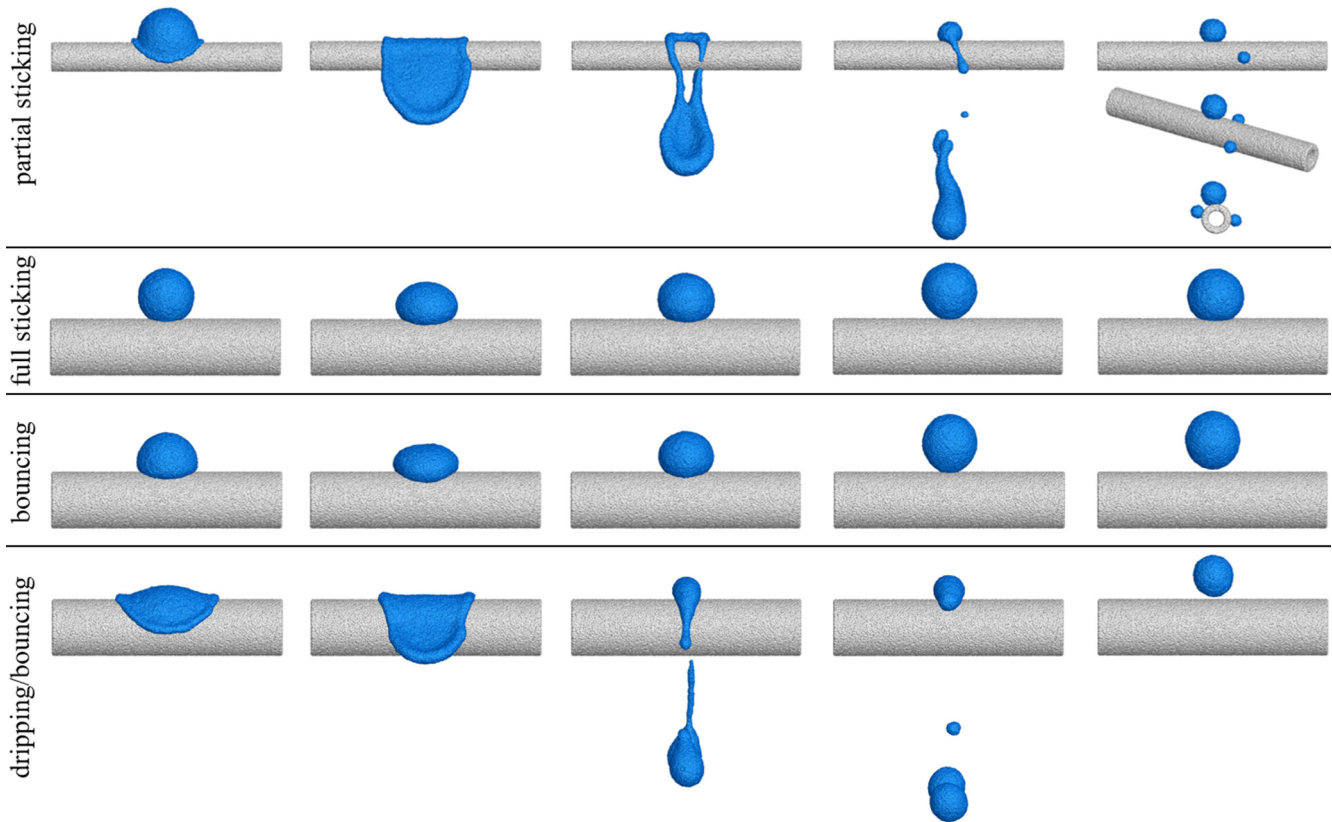


Fig. 6. Demonstrations of the four detachment modes, from upper to lower: partial sticking mode, full sticking mode, bouncing mode, and dripping/bouncing mode. For the last-line snapshots in partial sticking mode, three different views are given to show the partial sticking.

momentum during the retracting process. However, the spreading in azimuthal direction is very tricky: if the cylinder has a big size, the surface will be flatter than that on small cylinders, then the contact lines in azimuthal direction still have chance to retract and transfer the stored surface energy into an upward momentum; however, if the cylinder has a small size, the liquid within the spreading process in azimuthal direction is more likely to leave the cylinder surface and form a pendent lamella, or even a pendant secondary droplet at a relatively high impact velocity, and once the pendent secondary droplet appears, this part of liquid will have no chance to retract and transfer the surface energy into upward momentum as liquid retracting in the axial direction does, instead, they will drip from the cylinder, leading to the occurrence of dripping/bouncing mode.

Contact time is an important consideration in the design of anti-icing and self-cleaning surfaces, to reduce the contact time is always a goal that researchers pursue [35–40]. In the present work, contact time is defined as the time period from the moment of the droplet first touching the cylinder surface to the moment of the whole droplet leaving the surface (no liquid residues on the cylinder). It is also worthy to mention that for dripping/bouncing mode, the contact time includes the time for the part of the droplet on top to bounce off. Fig. 7 compares all the contact time for bouncing and dripping/bouncing cases mentioned in Fig. 5. Two distinct trends of the contact time variation can be observed: for all cylinders, in bouncing mode, the contact time increases with the increase of impact velocity, except a bouncing case on the small cylinder with $D^* = 0.5$ at $v = 1$; while in dripping/bouncing mode, the contact time decreases with the increase of impact velocity. Though it is well-known that the contact time of a droplet bouncing on a flat superhydrophobic surface is independent of the initial impact velocity [35], Gauthier et al. [38] and Guo et al. [39] show that

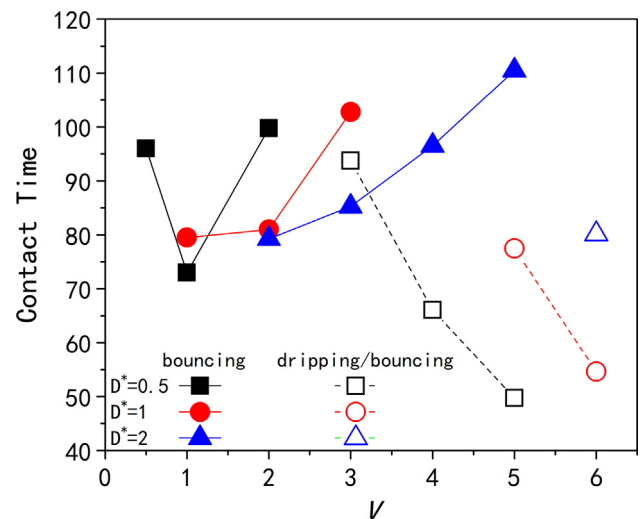


Fig. 7. Comparison of contact time on different cylinders at different initial impact velocity. Solid symbols denote the bouncing mode, hollow symbols denote the dripping/bouncing mode.

the contact time can be a function of the initial impact velocity when adding a cylindrical ridge on the flat surface: in both of their experimental studies, the contact time can decrease with the increase of the initial impact velocity. The present simulations further show that the contact time could be a more complex function of the initial impact velocity when using a pendant cylinder (which has no a supporting flat substrate like in Ref. 38 and Ref. 39) to replace the previous flat surfaces [35] or flat surfaces with cylindrical structures [38,39].

For bouncing mode, the unusual contact time dependence on impact velocity can be explained by the mismatch of spreading and retracting processes between axial and azimuthal directions, as shown in Fig. 8 which presents the time-dependent non-dimensional spreading diameters in both directions for impact on the cylinder with $D^* = 2$. For each case, especially cases with higher initial impact velocity, the movements of contact lines in the two directions are not at the same pace. Take the $v = 5$ case as an example, when the contact line in the axial direction goes into its retracting process after spreading to its maximum length, the contact line in the azimuthal direction is still spreading, this can be seen as a convergence stage reported in Zhang et al.'s work [12]. When the contact line in the axial direction retracts to a very narrow length, the droplet is elongated into a long, thin, and curving shape around the cylinder, as shown by the snapshot at $t = 29.4$ in Fig. 8. Due to the existence of surface tension and incompressibility of liquid, the contact line in the axial direction can hardly retract anymore and has to wait for the retraction of contact line from the azimuthal direction. Therefore, following a quick and short retracting process, there is a “waiting period” in the retracting process in the axial direction, as indicated by the yellow line in Fig. 8. During the “waiting period”, the spreading length in the axial direction keeps almost the same, as the snapshots at six moments within the period show, while the contact line in the azimuthal direction is still retracting. Only when the contact line in the azimuthal direction retracts to a certain length, the retracting in the axial direction can start again, and then along with retracting from both directions, the droplet can bounce off from the cylinder surface eventually. Since higher initial impact velocity can drive the contact line to spread further, and according to Fig. 8, retracting rates in the azimuthal direction indicated by the slopes of the retracting profiles, are almost the same, especially at higher impact velocities. Thus, the longer spreading length will take more time to retract back. Therefore, higher impact velocity which drives the contact line to spread further in the azimuthal direction will finally lead to longer contact time.

For dripping/bouncing mode, though there is also a mismatch of spreading and retracting processes between axial and azimuthal

directions, the contact time variation trend is opposite to that of the bouncing mode, thus, the underlying mechanism is also different. Fig. 9 shows the time-dependent non-dimensional spreading diameter in the azimuthal and axial directions for dripping/bouncing cases on $D^* = 0.5$ cylinder. Because of the existence of dripping in the retracting process in the azimuthal direction, there is a “necking period” from the beginning of the shrinking of the two lamellae to the dripping point. The dripping point is defined as the moment when the secondary droplets totally drip from the main droplet. The typical “necking period” of $v = 3$ impact case and the dripping points of every case can be found in Fig. 9, the snapshots of the droplet morphology at five different moments within the two periods can also be found in the figure. The “necking period” can extend the retracting process in the azimuthal direction, which can, in turn, extend the “waiting period” in the axial direction. The length of the “necking period” depends on the initial impact velocity, as higher impact velocity brings more downward momentum which can drive a faster dripping, or a shorter “necking period”, and finally leading to a shorter total contact time. In conclusion, in dripping/bouncing mode, the contact time increases with the decrease of initial impact velocity, this is opposite to the trend in bouncing mode. Though the contact time variation trend is the same with results in some previous studies [38,39], the underlying mechanisms are different due to different impact conditions. In their studies, the addition of a raised ridge with proper size on a flat surface can divide the initial droplet into several smaller parts, and smaller droplets need less time for bouncing off, therefore, the total contact time is reduced.

At last, another interesting effect caused by the different cylinder sizes is presented: the variation of splashing angle. When the droplet impact on the cylinders at an extremely high velocity, for instance, $v = 12$, splashing can happen, the upper snapshots in Fig. 10 show a typical splashing process. The splashing happens at the first moment when the droplet contacts the cylinders, after that, the droplet fractures and forms two net-like lamellae at the two sides. The two net-like lamellae can form an intersection angle and keep this angle for a while until the net-like lamellae further break up into many tiny droplets. This intersection angle is termed

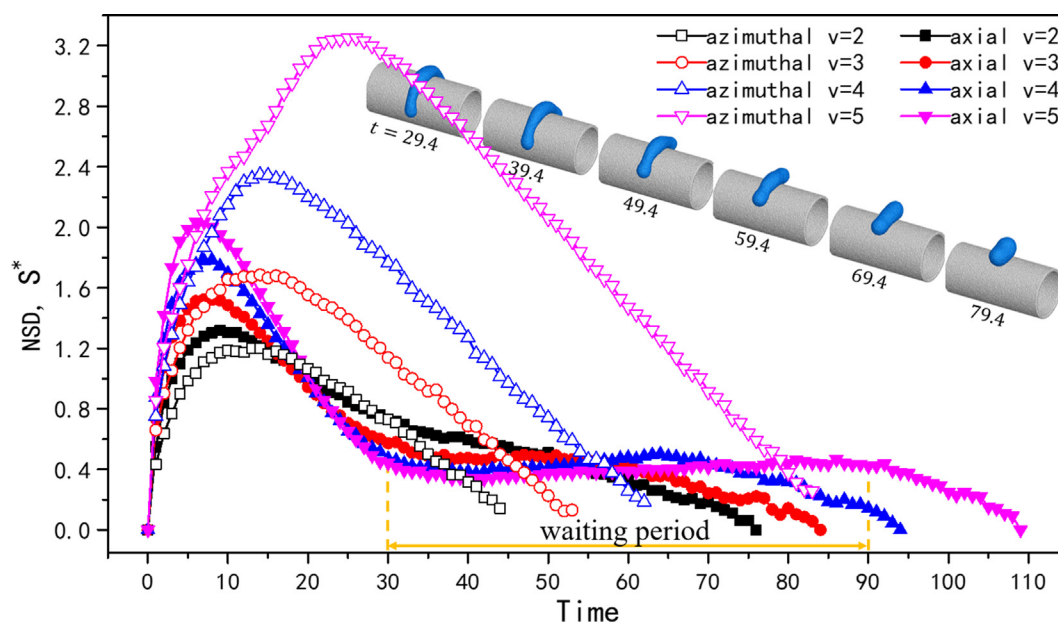


Fig. 8. Time-dependent non-dimensional spreading diameter (NSD, S^*) in the azimuthal and axial directions for bouncing cases on $D^* = 2$ cylinder. The yellow line indicates the “waiting period” of $v = 5$ impact case and the snapshots show the morphology of droplet at six moments within the “waiting period”. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

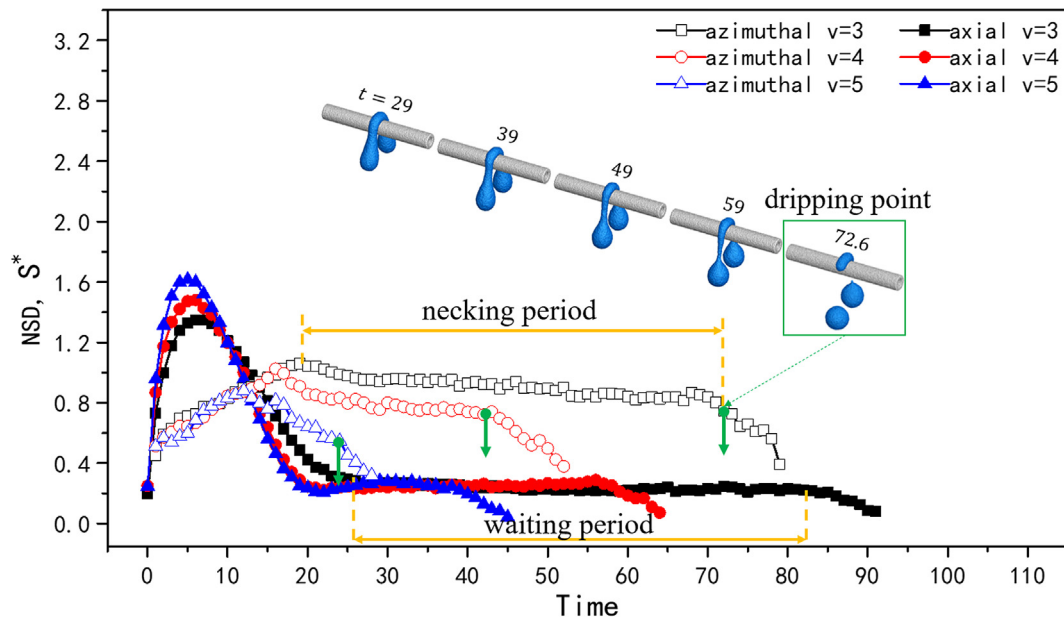


Fig. 9. Time-dependent non-dimensional spreading diameter in the azimuthal and axial directions for dripping/bouncing cases on $D^* = 0.5$ cylinder. The two horizontal yellow lines indicate the “waiting period” and “necking period” of $v = 3$ impact case in both directions, respectively. The snapshots show the morphology of droplet at five moments within the two periods. The green vertical arrows indicate the dripping points of the three cases. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

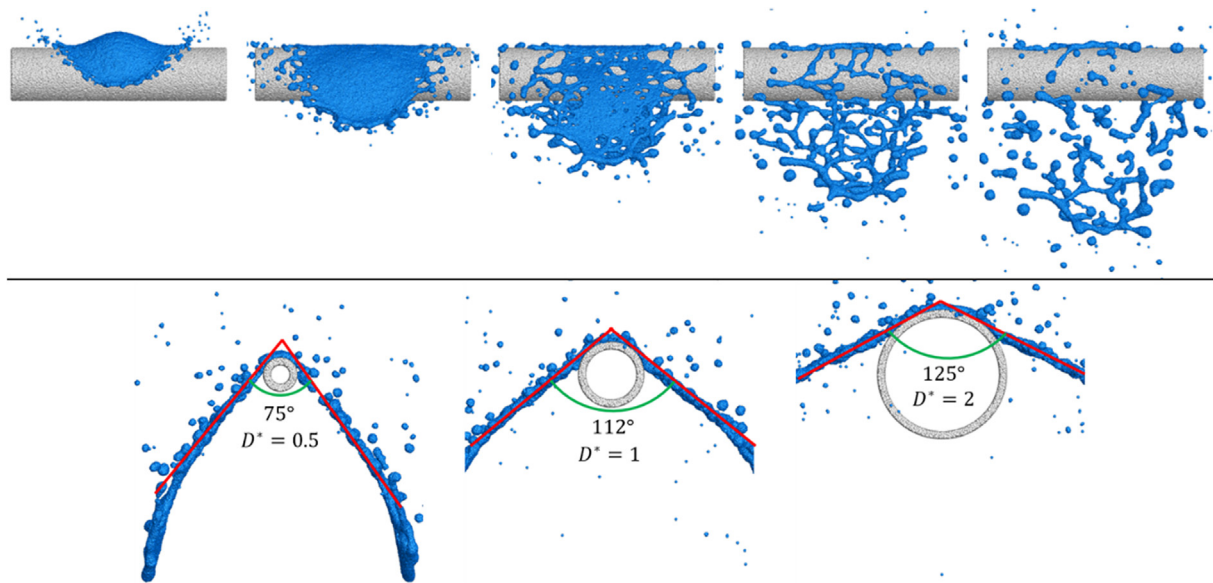


Fig. 10. Illustrations of a typical splashing process on the cylinder with $D^* = 1$ at $v = 12$ (upper snapshots) and different splashing angles on different cylinders at $v = 12$ (lower snapshots).

as splashing angle in the present work. What is surprising is that this splashing angle is a strong function of the cylinder size, as large cylinders can induce a larger splashing angle, as the lower snapshots in Fig. 10 show. The combined effect of the cylinder size and high impact velocity is attributed to this difference, as at high impact velocity, the contact line progressing direction is almost only dominated by the surface curve of the cylinder and the strong inertia at the very beginning of the impact, thus the relatively flat surface of large cylinders can guide the contact line to spread and leave the surface with a larger splashing angle, while relatively curve surface of a small cylinder will guide the contact line to spread and leave the surface with a smaller splashing angle.

4. Conclusions

The present work presents a numerical study of droplet impact on cylinders, and the influences of surface wettability, cylinder size, and impact velocity on the impact dynamics are studied. The simulation results show strong dependence on the droplet impact dynamics on them. Compared to previous studies which investigated the morphology of the droplet on cylindrical surfaces [8,12], the present study identifies three different droplet dripping morphologies, which are highly dependent on the surface wettability. The simulation results also show that the maximum non-dimensional spreading diameter in the axial direction is to be

found a strong function of the cylinder size and initial impact velocity, as it increases along with the increase of either the cylinder size or the impact velocity. However, the maximum wrapping angle on all cylinders will first increase and then decrease along with the increase of impact velocity. In the previous study, the contact time on a flat surface is found to be independent of the impact velocity [35], or the contact time can decrease with the increase of impact velocity on flat surface decorated with a ridge [38,39]. However, for the first time, our study shows a more complex contact time variation trend with initial impact velocity: it is not a monotonic function of the impact velocity, it can increase first with the increase of impact velocity, then decrease with the further increase of impact velocity. Our study further reveals that the “waiting period” and “necking period” in different detachment modes of the droplet can be attributed to the unusual variation trend. In bouncing mode, because the retracting rates in the azimuthal direction are always the same at different impact velocity, as higher impact velocity drives the contact line to spread further, more time is thus needed for its retraction, and the “waiting period” in the retracting process in the axial direction is therefore extended, leading to a longer contact time. In dripping/bouncing mode, because of the existence of dripping, the retracting process in the azimuthal direction is impeded, and a “necking period” is formed. As higher impact velocity which brings more downward momentum can speed up the dripping process, the “necking period” is thus reduced, and accordingly the “waiting period” is also reduced, leading to a shorter contact time. Also, in this simulation study, the splashing angle is found to depend on the cylinder size, as a large cylinder can induce a large splashing angle. The findings and understandings of the influences from surface wettability, cylinder size, and impact velocity on the droplet impact dynamics are expected to provide some guidance for the optimal design of anti-icing and self-cleaning surfaces in relevant applications, such as the power transmission line. However, there are still several unclear issues in the present work which need more effort in the future. For example: why the retracting rates in the azimuthal direction for different initial impact velocity are almost the same? How is the energy transition (surface energy to kinetic energy and vice versa) and distribution during the spreading and receding processes? And what are the underlying mechanisms of the splashing angle dependence on cylinder size? The answer to these questions would significantly contribute to the further understanding of the impact dynamics. Besides, in the future, the temperature should be considered in the impact process to investigate how the frozen solid surfaces influence the spreading and retracting processes, and how to reduce the contact time before icing.

CRediT authorship contribution statement

Yulei Wang: Methodology, Software, Formal analysis, Investigation, Data curation, Visualization. **Yuxiang Wang:** Conceptualization, Writing - original draft, Supervision. **Shaorong Wang:** Investigation, Data curation, Visualization, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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